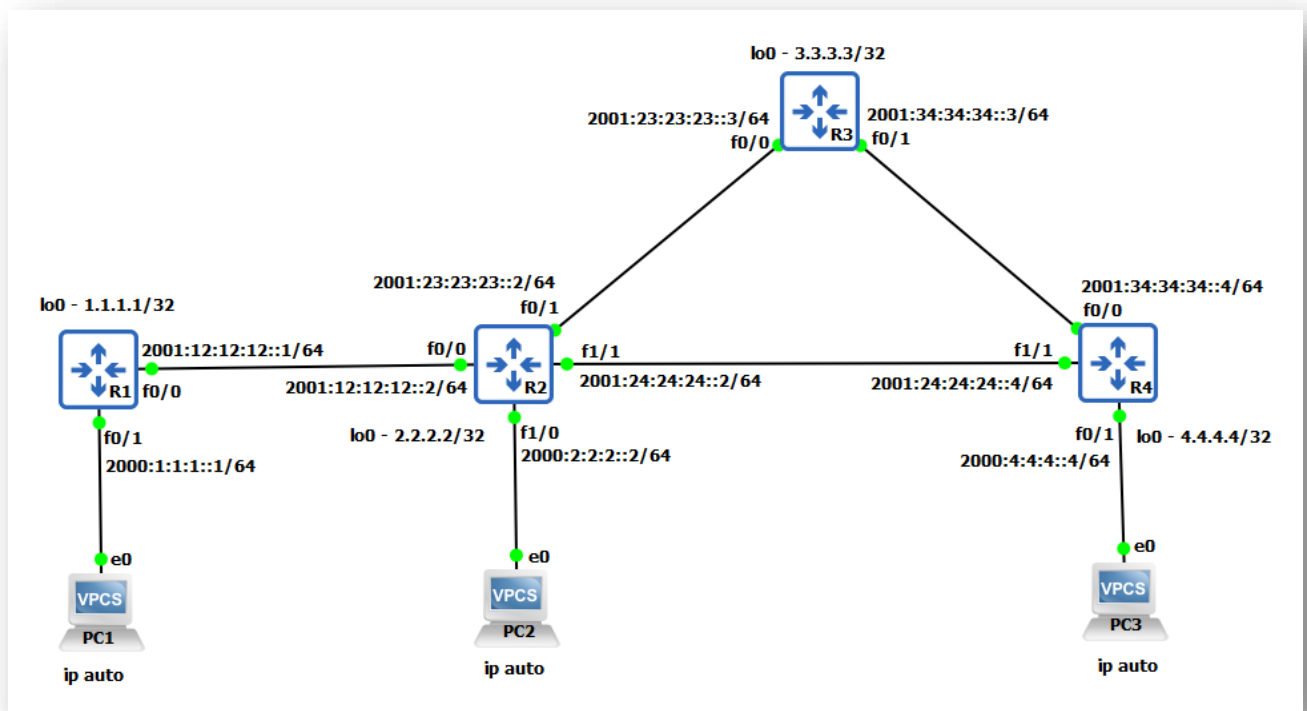


EIGRP - IPv6



Task

1. Configure routers with IPv6 address as shown in topology and configure enable password as **ccna** & device access password as **cisco**.
2. Configure EIGRPv6 process with ASN 100 on all routers and advertise its directly connected networks.
3. All router should be able to ping each other's PCs IP.
4. PCs should be able to ping each other's IP.

Task-1: Router configuration with IPv6

R1 Router Configuration:

```

R1#enable
R1#config t
R1(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#ipv6 unicast-routing
R1(config)#int fa0/1
R1(config-if)#ipv6 address 2000:1:1:1::1/64
R1(config-if)#description Link to Local LAN
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int fa0/0
R1(config-if)#ipv6 address 2001:12:12:12::1/64

```

```

R1(config-if)#description Link to R2 on fa0/0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int lo0
R1(config-if)#ip address 1.1.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#line console 0
R1(config-line)#password cisco1
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 0 0
R1(config-line)#exit
R1(config)#exit
R1#

```

R2 Router Configuration:

```

R2#enable
R2#config t
R2(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable password ccna
R2(config)#ipv6 unicast-routing
R2(config)#int fa1/0
R2(config-if)#ipv6 address 2000:2:2:2::2/64
R2(config-if)#description Link to Local LAN
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa1/1
R2(config-if)#ipv6 address 2001:24:24:24::2/64
R2(config-if)#description Link to R4 on fa1/1
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa0/1
R2(config-if)#ipv6 address 2001:23:23:23::2/64
R2(config-if)#description Link to R3 on fa0/0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa0/0
R2(config-if)#ipv6 address 2001:12:12:12::2/64
R2(config-if)#description Link to R1 on fa0/0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int lo0
R2(config-if)#ip address 2.2.2.2 255.255.255.0

```

```
R2(config-if)#exit
R2(config)#line vty 0 15
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 1 0
R2(config-line)#exit
R2(config)#line console 0
R2(config-line)#password cisco1
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 0 0
R2(config-line)#exit
R2(config)#exit
R2#
```

R3 Router Configuration:

```
R3#enable
R3#config t
R3(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable password ccna
R3(config)#ipv6 unicast-routing
R3(config)#int fa0/0
R3(config-if)#ipv6 address 2001:23:23:23::3/64
R3(config-if)#description Link to R2 on fa0/1
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int fa0/1
R3(config-if)#ipv6 address 2001:34:34:34::3/64
R3(config-if)#description Link to R4 on fa0/0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int lo0
R3(config-if)#ip address 3.3.3.3 255.255.255.0
R3(config-if)#exit
R3(config)#line vty 0 15
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 1 0
R3(config-line)#exit
R3(config)#line console 0
R3(config-line)#password cisco1
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 0 0
R3(config-line)#exit
R3(config)#exit
R3#
```

R4 Router Configuration:

```

R4#config t
R4(config)#hostname R4
R4(config)#no ip domain-lookup
R4(config)#enable password ccna
R4(config)#ipv6 unicast-routing
R4(config)#int fa0/0
R4(config-if)#ipv6 address 2001:34:34:34::4/64
R4(config-if)#description Link to R3 on fa0/1
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int fa0/1
R4(config-if)#ipv6 address 2000:4:4:4::4/64
R4(config-if)#description Link to local LAN
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int fa1/1
R4(config-if)#ipv6 address 2001:24:24:24::4/64
R4(config-if)#description Link to R5 on fa0/0
R4(config-if)#no shut
R4(config-if)#exit
R4(config)#int lo0
R4(config-if)#ip address 4.4.4.4 255.255.255.0
R4(config-if)#exit
R4(config)#line vty 0 15
R4(config-line)#password cisco
R4(config-line)#login
R4(config-line)#logging synchronous
R4(config-line)#exec-timeout 1 0
R4(config-line)#exit
R4(config)#line console 0
R4(config-line)#password cisco1
R4(config-line)#login
R4(config-line)#logging synchronous
R4(config-line)#exec-timeout 0 0
R4(config-line)#exit
R4(config)#exit
R4#

```

PC1 Configuration:

```

PC1> ip auto
GLOBAL SCOPE      : 2000:1:1:1:2050:79ff:fe66:6800/64
ROUTER LINK-LAYER : ca:01:20:c0:00:06

```

PC2 Configuration:

```

PC2> ip auto
GLOBAL SCOPE      : 2000:2:2:2:2050:79ff:fe66:6801/64
ROUTER LINK-LAYER : ca:02:33:d4:00:1c

```

PC3 Configuration:

```
PC3> ip auto
GLOBAL SCOPE : 2000:4:4:4:2050:79ff:fe66:6802/64
ROUTER LINK-LAYER : ca:04:29:78:00:06
```

✓ Verification & Testing:**R1#sh ipv6 interface brief | exclude unassigned|down**

```
FastEthernet0/0 [up/up]
FE80::C801:20FF:FEC0:8
2001:12:12:12::1
FastEthernet0/1 [up/up]
FE80::C801:20FF:FEC0:6
2000:1:1:1::1
Loopback0 [up/up]
```

R2#sh ipv6 interface brief | exclude unassigned|down

```
FastEthernet0/0 [up/up]
FE80::C802:33FF:FED4:8
2001:12:12:12::2
FastEthernet0/1 [up/up]
FE80::C802:33FF:FED4:6
2001:23:23:23::2
FastEthernet1/0 [up/up]
FE80::C802:33FF:FED4:1C
2000:2:2:2::2
FastEthernet1/1 [up/up]
FE80::C802:33FF:FED4:1D
2001:24:24:24::2
Loopback0 [up/up]
```

R3#sh ipv6 interface brief | exclude unassigned|down

```
FastEthernet0/0 [up/up]
FE80::C803:1AFF:FE28:8
2001:23:23:23::3
FastEthernet0/1 [up/up]
FE80::C803:1AFF:FE28:6
2001:34:34:34::3
Loopback0 [up/up]
```

R4#sh ipv6 interface brief | exclude unassigned|down

```
FastEthernet0/0 [up/up]
FE80::C804:29FF:FE78:8
2001:34:34:34::4
FastEthernet0/1 [up/up]
FE80::C804:29FF:FE78:6
2000:4:4:4::4
FastEthernet1/1 [up/up]
FE80::C804:29FF:FE78:1D
2001:24:24:24::4
```

Loopback0 [up/up]

Task2 : Configure EIGRPv6 process with ASN 100

EIGRP Configuration on R1:

```
R1#config t
R1(config)#ipv6 router eigrp 100
R1(config-rtr)#eigrp router-id 1.1.1.1
R1(config-rtr)#no shut
R1(config-rtr)#exit
R1(config)#int fa0/0
R1(config-if)#ipv6 eigrp 100
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ipv6 eigrp 100
R1(config-if)#exit
R1(config)#exit
R1#
```

EIGRP Configuration on R2:

```
R2#config t
R2(config)#ipv6 router eigrp 100
R2(config-rtr)#eigrp router-id 2.2.2.2
R2(config-rtr)#no shut
R2(config-rtr)#exit
R2(config)#int fa0/0
R2(config-if)#ipv6 eigrp 100
R2(config-if)#exit
R2(config)#int fa0/1
R2(config-if)#ipv6 eigrp 100
R2(config-if)#exit
R2(config)#int fa1/1
R2(config-if)#ipv6 eigrp 100
R2(config-if)#exit
R2(config)#int fa1/0
R2(config-if)#ipv6 eigrp 100
R2(config-if)#exit
R2(config)#exit
R2#
```

EIGRP Configuration on R3:

```
R3#config t
R3(config)#ipv6 router eigrp 100
R3(config-rtr)#eigrp router-id 3.3.3.3
R3(config-rtr)#no shut
R3(config-rtr)#exit
R3(config)#int fa0/0
R3(config-if)#ipv6 eigrp 100
```

```
R3(config-if)#exit
R3(config)#int fa0/1
R3(config-if)#ipv6 eigrp 100
R3(config-if)#exit
R3(config)#exit
R3#
```

EIGRP Configuration on R4:

```
R4#config t
R4(config)#ipv6 router eigrp 100
R4(config-rtr)#eigrp router-id 4.4.4.4
R4(config-rtr)#no shut
R4(config-rtr)#exit
R4(config)#int fa0/0
R4(config-if)#ipv6 eigrp 100
R4(config-if)#exit
R4(config)#int fa0/1
R4(config-if)#ipv6 eigrp 100
R4(config-if)#exit
R4(config)#int fa1/1
R4(config-if)#ipv6 eigrp 100
R4(config-if)#exit
R4(config)#exit
R4#
```

✓ Verification & Testing:

R1#sh ipv6 protocols

IPv6 Routing Protocol is "connected"

IPv6 Routing Protocol is "ND"

IPv6 Routing Protocol is "eigrp 100"

EIGRP-IPv6 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 1.1.1.1

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 16

Maximum hopcount 100

Maximum metric variance 1

Interfaces:

FastEthernet0/0

FastEthernet0/1

Redistribution:

None

R1#

R2#sh ipv6 protocols

IPv6 Routing Protocol is "connected"

IPv6 Routing Protocol is "ND"

IPv6 Routing Protocol is "eigrp 100"

EIGRP-IPv6 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 2.2.2.2

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 16

Maximum hopcount 100

Maximum metric variance 1

Interfaces:

FastEthernet0/0

FastEthernet0/1

FastEthernet1/1

FastEthernet1/0

Redistribution:

None

R2#

R3#sh ipv6 protocols

IPv6 Routing Protocol is "connected"

IPv6 Routing Protocol is "ND"

IPv6 Routing Protocol is "eigrp 100"

EIGRP-IPv6 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 3.3.3.3

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 16

Maximum hopcount 100

Maximum metric variance 1

Interfaces:

FastEthernet0/0

FastEthernet0/1

Redistribution:

None

R3#

R4#sh ipv6 protocols

IPv6 Routing Protocol is "connected"

IPv6 Routing Protocol is "ND"

IPv6 Routing Protocol is "eigrp 100"

EIGRP-IPv6 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 4.4.4.4

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 16

Maximum hopcount 100

Maximum metric variance 1

Interfaces:

FastEthernet0/0

FastEthernet0/1

FastEthernet1/1

Redistribution:

None

R4#

R1#sh ipv6 eigrp neighbors

EIGRP-IPv6 Neighbors for AS(100)

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
0	Link-local address: FE80::C802:33FF:FED4:8	Fa0/0	12	00:05:39	26	156	0	11

R1#

R2#sh ipv6 eigrp neighbors

EIGRP-IPv6 Neighbors for AS(100)

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
2	Link-local address: FE80::C804:29FF:FE78:1D	Fa1/1	13	00:05:13	34	204	0	6
1	Link-local address: FE80::C803:1AFF:FE28:8	Fa0/1	14	00:05:26	36	216	0	6
0	Link-local address: FE80::C801:20FF:FEC0:8	Fa0/0	13	00:05:43	1026	5000	0	5

R2#

R3#sh ipv6 eigrp neighbors

EIGRP-IPv6 Neighbors for AS(100)

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
1	Link-local address: FE80::C804:29FF:FE78:8	Fa0/1	14	00:05:17	46	276	0	5
0	Link-local address: FE80::C802:33FF:FED4:6	Fa0/0	14	00:05:30	1270	5000	0	10

R3#

R4#sh ipv6 eigrp neighbors

EIGRP-IPv6 Neighbors for AS(100)

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
1	Link-local address: FE80::C802:33FF:FED4:1D	Fa1/1	12	00:05:19	1573	5000	0	12

0 Link-local address: Fa0/0 12 00:05:19 1577 5000 0 7
 FE80::C803:1AFF:FE28:6
 R4#

R1#sh ipv6 route eigrp

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, **D - EIGRP**, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

D 2000:2:2:2::/64 [90/30720]

via FE80::C802:33FF:FED4:8, FastEthernet0/0

D 2000:4:4:4::/64 [90/33280]

via FE80::C802:33FF:FED4:8, FastEthernet0/0

D 2001:23:23:23::/64 [90/30720]

via FE80::C802:33FF:FED4:8, FastEthernet0/0

D 2001:24:24:24::/64 [90/30720]

via FE80::C802:33FF:FED4:8, FastEthernet0/0

D 2001:34:34:34::/64 [90/33280]

via FE80::C802:33FF:FED4:8, FastEthernet0/0

R1#

R2#sh ipv6 route eigrp

IPv6 Routing Table - default - 12 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, **D - EIGRP**, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, I - LISP

D 2000:1:1:1::/64 [90/30720]

via FE80::C801:20FF:FEC0:8, FastEthernet0/0

D 2000:4:4:4::/64 [90/30720]

via FE80::C804:29FF:FE78:1D, FastEthernet1/1

D 2001:34:34:34::/64 [90/30720]

via FE80::C803:1AFF:FE28:8, FastEthernet0/1

via FE80::C804:29FF:FE78:1D, FastEthernet1/1

R2#

R3#sh ipv6 route eigrp

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, **D - EIGRP**, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

```

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP
D 2000:1:1:1::/64 [90/33280]
  via FE80::C802:33FF:FED4:6, FastEthernet0/0
D 2000:2:2:2::/64 [90/30720]
  via FE80::C802:33FF:FED4:6, FastEthernet0/0
D 2000:4:4:4::/64 [90/30720]
  via FE80::C804:29FF:FE78:8, FastEthernet0/1
D 2001:12:12:12::/64 [90/30720]
  via FE80::C802:33FF:FED4:6, FastEthernet0/0
D 2001:24:24:24::/64 [90/30720]
  via FE80::C802:33FF:FED4:6, FastEthernet0/0
  via FE80::C804:29FF:FE78:8, FastEthernet0/1
R3#

```

R4#sh ipv6 route eigrp

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP

H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea

IS - ISIS summary, **D - EIGRP**, EX - EIGRP external, NM - NEMO

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP

```

D 2000:1:1:1::/64 [90/33280]
  via FE80::C802:33FF:FED4:1D, FastEthernet1/1
D 2000:2:2:2::/64 [90/30720]
  via FE80::C802:33FF:FED4:1D, FastEthernet1/1
D 2001:12:12:12::/64 [90/30720]
  via FE80::C802:33FF:FED4:1D, FastEthernet1/1
D 2001:23:23:23::/64 [90/30720]
  via FE80::C803:1AFF:FE28:6, FastEthernet0/0
  via FE80::C802:33FF:FED4:1D, FastEthernet1/1
R4#

```

R1#sh ipv6 eigrp topology

EIGRP-IPv6 Topology Table for AS(100)/ID(1.1.1.1)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - reply Status, s - sia Status

```

P 2001:24:24:24::/64, 1 successors, FD is 30720
  via FE80::C802:33FF:FED4:8 (30720/28160), FastEthernet0/0
P 2001:12:12:12::/64, 1 successors, FD is 28160
  via Connected, FastEthernet0/0
P 2000:4:4:4::/64, 1 successors, FD is 33280
  via FE80::C802:33FF:FED4:8 (33280/30720), FastEthernet0/0
P 2001:34:34:34::/64, 1 successors, FD is 33280
  via FE80::C802:33FF:FED4:8 (33280/30720), FastEthernet0/0
P 2000:2:2:2::/64, 1 successors, FD is 30720
  via FE80::C802:33FF:FED4:8 (30720/28160), FastEthernet0/0
P 2000:1:1:1::/64, 1 successors, FD is 28160
  via Connected, FastEthernet0/1

```

P 2001:23:23:23::/64, 1 successors, FD is 30720
via FE80::C802:33FF:FED4:8 (30720/28160), FastEthernet0/0

R2#sh ipv6 eigrp topology

EIGRP-IPv6 Topology Table for AS(100)/ID(2.2.2.2)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - reply Status, s - sia Status

P 2001:24:24:24::/64, 1 successors, FD is 28160
via Connected, FastEthernet1/1

P 2001:12:12:12::/64, 1 successors, FD is 28160
via Connected, FastEthernet0/0

P 2000:4:4:4::/64, 1 successors, FD is 30720
via FE80::C804:29FF:FE78:1D (30720/28160), FastEthernet1/1

P 2001:34:34:34::/64, 2 successors, FD is 30720
via FE80::C803:1AFF:FE28:8 (30720/28160), FastEthernet0/1
via FE80::C804:29FF:FE78:1D (30720/28160), FastEthernet1/1

P 2000:1:1:1::/64, 1 successors, FD is 30720
via FE80::C801:20FF:FEC0:8 (30720/28160), FastEthernet0/0

P 2000:2:2:2::/64, 1 successors, FD is 28160
via Connected, FastEthernet1/0

P 2001:23:23:23::/64, 1 successors, FD is 28160
via Connected, FastEthernet0/1

R3#sh ipv6 eigrp topology

EIGRP-IPv6 Topology Table for AS(100)/ID(3.3.3.3)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - reply Status, s - sia Status

P 2001:24:24:24::/64, 2 successors, FD is 30720
via FE80::C802:33FF:FED4:6 (30720/28160), FastEthernet0/0
via FE80::C804:29FF:FE78:8 (30720/28160), FastEthernet0/1

P 2001:12:12:12::/64, 1 successors, FD is 30720
via FE80::C802:33FF:FED4:6 (30720/28160), FastEthernet0/0

P 2000:4:4:4::/64, 1 successors, FD is 30720
via FE80::C804:29FF:FE78:8 (30720/28160), FastEthernet0/1

P 2001:34:34:34::/64, 1 successors, FD is 28160
via Connected, FastEthernet0/1

P 2000:1:1:1::/64, 1 successors, FD is 33280
via FE80::C802:33FF:FED4:6 (33280/30720), FastEthernet0/0

P 2000:2:2:2::/64, 1 successors, FD is 30720
via FE80::C802:33FF:FED4:6 (30720/28160), FastEthernet0/0

P 2001:23:23:23::/64, 1 successors, FD is 28160
via Connected, FastEthernet0/0

R4#sh ipv6 eigrp topology

EIGRP-IPv6 Topology Table for AS(100)/ID(4.4.4.4)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - reply Status, s - sia Status

P 2001:24:24:24::/64, 1 successors, FD is 28160

```

via Connected, FastEthernet1/1
P 2001:12:12:12::/64, 1 successors, FD is 30720
  via FE80::C802:33FF:FED4:1D (30720/28160), FastEthernet1/1
P 2000:4:4:4::/64, 1 successors, FD is 28160
  via Connected, FastEthernet0/1
P 2001:34:34:34::/64, 1 successors, FD is 28160
  via Connected, FastEthernet0/0
P 2000:1:1:1::/64, 1 successors, FD is 33280
  via FE80::C802:33FF:FED4:1D (33280/30720), FastEthernet1/1
P 2000:2:2:2::/64, 1 successors, FD is 30720
  via FE80::C802:33FF:FED4:1D (30720/28160), FastEthernet1/1
P 2001:23:23:23::/64, 2 successors, FD is 30720
  via FE80::C802:33FF:FED4:1D (30720/28160), FastEthernet1/1
  via FE80::C803:1AFF:FE28:6 (30720/28160), FastEthernet0/0

```

R1#sh ipv6 eigrp 100 interfaces detail fa0/0

EIGRP-IPv6 Interfaces for AS(100)

Interface	Peers	Xmit Queue Un/Reliable	PeerQ Un/Reliable	Mean SRTT	Pacing Time Un/Reliable	Multicast Flow Timer	Pending Routes
Fa0/0	1	0/0	0/0	26	0/0	96	0

Hello-interval is 5, Hold-time is 15

Split-horizon is enabled

Next xmit serial <none>

Packetized sent/expedited: 3/0

Hello's sent/expedited: 1214/2

Un/reliable mcasts: 0/4 Un/reliable ucasts: 4/2

Mcast exceptions: 0 CR packets: 0 ACKs suppressed: 0

Retransmissions sent: 1 Out-of-sequence rcvd: 1

Topology-ids on interface - 0

Authentication mode is not set

R1#

Ping PC3 and PC2 from R1

R1#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 28/47/116 ms

R1#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:
!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/56/64 ms

Ping PC1 and PC3 from R2

R2#ping ipv6 2000:1:1:1:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:2050:79FF:FE66:6800, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/39/88 ms

R2#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/33/68 ms

Ping PC1, PC2 and PC3 from R3

R3#ping ipv6 2000:1:1:1:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:2050:79FF:FE66:6800, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 28/48/64 ms

R3#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/25/32 ms

R3#ping ipv6 2000:4:4:4:2050:79ff:fe66:6802

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:4:4:4:2050:79FF:FE66:6802, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/45/128 ms

Ping PC1 and PC2 from R4

R4#ping ipv6 2000:1:1:1:2050:79ff:fe66:6800

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:1:1:1:2050:79FF:FE66:6800, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 28/54/92 ms

R4#ping ipv6 2000:2:2:2:2050:79ff:fe66:6801

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2000:2:2:2:2050:79FF:FE66:6801, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/32/68 ms

Ping PC2 & PC3 from PC1

PC1> ping 2000:2:2:2:2050:79ff:fe66:6801

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=1 ttl=60 time=40.874 ms

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=2 ttl=60 time=42.183 ms

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=3 ttl=60 time=40.084 ms

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=4 ttl=60 time=42.165 ms

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=5 ttl=60 time=43.191 ms

PC1> ping 2000:4:4:4:2050:79ff:fe66:6802

2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=1 ttl=58 time=82.982 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=2 ttl=58 time=61.104 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=3 ttl=58 time=61.242 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=4 ttl=58 time=63.037 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=5 ttl=58 time=64.251 ms

Ping PC1 & PC3 from PC2

PC2> ping 2000:1:1:1:2050:79ff:fe66:6800

2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=1 ttl=60 time=48.040 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=2 ttl=60 time=43.106 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=3 ttl=60 time=42.231 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=4 ttl=60 time=42.192 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=5 ttl=60 time=42.175 ms

PC2> ping 2000:4:4:4:2050:79ff:fe66:6802

2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=1 ttl=60 time=53.013 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=2 ttl=60 time=41.084 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=3 ttl=60 time=43.157 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=4 ttl=60 time=41.013 ms
2000:4:4:4:2050:79ff:fe66:6802 icmp6_seq=5 ttl=60 time=41.134 ms

Ping PC1 & PC2 from PC3

PC3> ping 2000:1:1:1:2050:79ff:fe66:6800

2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=1 ttl=58 time=64.952 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=2 ttl=58 time=63.045 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=3 ttl=58 time=64.101 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=4 ttl=58 time=64.100 ms
2000:1:1:1:2050:79ff:fe66:6800 icmp6_seq=5 ttl=58 time=63.141 ms

PC3> ping 2000:2:2:2:2050:79ff:fe66:6801

2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=1 ttl=60 time=36.300 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=2 ttl=60 time=42.277 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=3 ttl=60 time=42.286 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=4 ttl=60 time=42.275 ms
2000:2:2:2:2050:79ff:fe66:6801 icmp6_seq=5 ttl=60 time=42.306 ms

Important Commands:

```
sh ipv6 interface brief | exclude unassigned|down  
sh ipv6 protocols  
sh ipv6 eigrp neighbors  
sh ipv6 route eigrp  
sh ipv6 eigrp topology  
sh ipv6 eigrp 100 interfaces detail fa/x
```

Ratan